

DID THE DODD-FRANK ACT IMPACT PRIVATE FUND PERFORMANCE ? – EVIDENCE FROM
2010 – 2015

WULF A. KAAL (CORRESPONDING AUTHOR)

Prof. Wulf A. Kaal, Ph.D.
Associate Professor
University of St. Thomas School of Law
1000 LaSalle Avenue
Minneapolis, MN 55403
USA
Cell: (312) 810-4390
Email: wulfkaal@stthomas.edu

BARBARA LUPPI

Prof. Barbara Luppi, Ph.D.
University of Modena and Reggio Emilia
Department of Economics
Viale J. Berengario 51,
41100 Modena
Italy
Tel: +39 059 2056845
E-mail: barbara.luppi@unimore.it

SANDRA PATERLINI

Prof. Sandra Paterlini, Ph.D.
Chair of Financial Econometrics & Asset Management
EBS Universität für Wirtschaft & Recht
EBS Business School
Department of Finance, Accounting and Real Estate
Gustav-Stresemann-Ring 3
D-65189 Wiesbaden
Germany
Phone: +49 611 7102 1227
Email: sandra.paterlini@ebs.edu

ABSTRACT

Title IV of the Dodd-Frank Act introduced the most significant regulatory change in the history of the private fund industry. To analyze the effect of Title IV on the private fund industry, we use five years of private fund performance data with over 7,000 reporting private funds. Our findings do not support the private fund industry's claims that increased supervision and disclosure mandated in the Dodd-Frank Act have a negative effect on private fund earnings.

Keywords: Dodd-Frank Act, Private Funds, Performance, Regression Discontinuity Design

JEL Classification: G23, G24, G28, K22

1. INTRODUCTION

Private fund managers fear that the registration and disclosure requirements under the Dodd-Frank Act could lower their returns (Oesterle 2006, Kaal 2013a, Kaulesar 2012). A survey of private fund managers conducted in 2012, after the registration effective date for private fund managers under the Dodd-Frank Act, revealed that a clear majority of private fund managers believed that increased compliance costs negatively affect the industry. However, while Dodd-Frank Act compliance costs affect the profitability of private fund advisors' investment management companies, the majority of private fund manager respondents in a leading survey on the issue opined that registration and disclosure requirements under the Dodd-Frank Act do not affect the returns of the private fund industry (Kaal 2013a).

For the first time since the inception of the private fund industry, Title IV of the Dodd-Frank Act (the Act or Dodd-Frank) and Securities and Exchange Commission (SEC) rules implementing the Act require private fund manager registration and enhanced disclosure of sensitive proprietary information (Dodd-Frank Act § 401, 402, 408). The SEC's rules introduced controversial disclosure obligations that require the reporting of risk metrics, counterparties and credit exposure, strategies and products used by the investment adviser and its funds, performance and changes in performance, financing information, positions held by the investment adviser, percentage of assets traded using algorithms, and the percentage of equity and debt, among others (SEC Form PF 2012, SEC Form ADV 2012). The true impact of these regulations on the private fund industry and their effect on private fund advisor performance is unclear. Some analysts estimate that the cost will range from \$50,000 to \$400,000 per year (Kaal 2013a and 2015b).

The enactment of Title IV of the Dodd-Frank Act was controversial. Despite reservations on both sides, the regulation of private funds in Title IV was included in the Dodd-Frank Act (CoE 2011, Paletta and Lucchetti 2010). In a 2009 white paper, the Treasury Department favored the registration of advisers of private funds and other private pools of capital with the SEC to ensure that financial institutions that are critical to market functioning are subject to strong oversight (Department of the Treasury, 2009). SEC Commissioner Louis Aguilar commented: "We're totally unable to discern what's going on in [the private fund] market, [we] have no idea how many dollars are involved, [. . .] what type of risk-taking is happening, [we] don't know if they're investing in vanilla securities or investing in the riskiest instruments." (Goldfarb and Cho, 2009). In the debate over the measure, Senator Jack Reed (D-RI) commented (2009, 1): "These private pools of capital are responsible for huge transfers of capital and risk, and so examining these industries and potential regulation are extremely important." Rep. Paul E. Kanjorski (D-PA) (2009, H14420) stated: "[F]or the first time regulators will have the information needed to better understand exactly how these entities operate and whether their actions pose a threat to the financial system as a whole." Opponents of Title IV were concerned that regulation would invade the privacy of clients and would place an unnecessary burden on investment advisers (Strasburg, 2009). Banking Committee ranking member Richard Shelby (R-Ala.), complained that the bill represents a 'squandered opportunity' for streamlining the regulatory system and predicted that the measure will foster 'unrestrained and unaccountable' agencies." (Ferullo, Bruce, Hill, and Manickavasgam 2010).

Did Congress by enacting Title IV of the Dodd-Frank Act overburden the private fund industry? This study aims to estimate the impact of private fund adviser registration

and increased disclosure requirements under the Dodd-Frank Act on private fund performance. We use self-reported Morningstar earnings data for 3,424 private fund advisers that are based in the United States and are subject to the registration and disclosure requirements under Title IV of the Dodd-Frank Act and SEC rules. Our dataset comprises earnings data for private fund advisers from 2010 to 2015.

We find mixed evidence but conclude that private fund adviser registration and disclosure under the Dodd-Frank Act had no significant effect on private fund adviser returns in March 2012. This evidence contradicts claims of the private fund industry that private fund adviser registration under the Dodd-Frank Act negatively affects private fund performance. Because the private fund industry is a significant representative of Wall Street's interests and regulating private funds is politically sensitive, the findings in this study are intended to provide guidance to policy makers who seek to implement a regulatory framework for the private fund industry.

Part II explains the core aspects of private fund registration and enhanced disclosure requirements under the Dodd-Frank Act and SEC implementation rules and provides a short overview of the literature on private fund performance. Part III describes our dataset and the empirical methodology. Part IV reports the empirical results of the impact of Title IV of the Dodd-Frank Act and SEC implementation rules on private fund performance, provides a summary of key findings, and discusses their implications, limitations, and the need for future research. Part V concludes.

2. PRIVATE FUND ADVISER REGULATION UNDER THE DODD-FRANK ACT

The private fund industry evolved without substantial regulatory oversight (Kaal 2009, Fung and Hsieh 1999). Since the development of the private fund model by Alfred Winslow Jones in the late 1940s (Jones 1949, Loomis 1966, Landau 1968), balancing the interests of private fund managers, investors, and regulators to attain a suitable level of regulatory oversight has proved contentious (Kaal 2013a). The SEC and the private fund industry reached a workable compromise in the mid 1980s, allowing private funds to remain exempt from regulation as long as they complied with accredited investor standards (SEC Regulation D, IA Release No. 2576, IA Release No. 2628) and safe harbor requirements (Kaal 2011 and 2013a).

The expansion of the private fund industry in the late 1990s and its increasing importance in financial markets in combination with the creation of global financial markets and the non-stop flow of new financial instruments (Schumer 2008) changed the delicate balance of interests. The retailization of the private fund industry (IA Release 2333, Kaal 2009, Donaldson 2003), the rise of private fund fraud (IA Release 2333), and the collapse of large private funds such as Long Term Capital Management in 1998, Tiger Funds in 2000, and Amaranth in 2006 (Roth and Fortune 2001, Jickling and Raab 2006, Jorion 2000), among others, precipitated politicians and policy makers to voice concern over the alleged systemic risks posed by the private fund industry (Cox 2006, Kaal 2009 and 2013a). A combination of these factors amplified calls for increased supervision of the private fund industry (SEC PRIVATE FUND REPORT 2003).

Following the global financial crisis of 2008-09, however, the tension between the private fund industry and the regulators flared up again. Politicians and the media accused private funds for taking undue risks that contributed to the financial crisis (Lee 2009, Economist 2010). Capitalizing on the political support for increased oversight of the

private fund industry, legislatures in the United States and Europe enacted new rules intended to address the perceived shortcomings of the global private fund industry (Dodd-Frank Act §§ 401-416, Commission AIF Proposal 2009). In the United States, Congress enacted the Private Fund Investment Advisers Registration Act of 2010 (PFIARA) in Title IV of the Dodd-Frank Act (Dodd-Frank Act §§ 401-416). Title IV was intended to close regulatory gaps, provide greater protections for investors, and curtail speculative trading practices (House of Representatives Joint Explanatory Statement 2010).

Title IV established rules and regulations for the registration of private funds with the SEC and expands the reporting requirements of private advisers to the SEC (Dodd-Frank Act § 408). The registration of private fund advisers was intended to limit systemic risk, prevent fraud, provide information to investors (Durbin 2010), and help the SEC to restrict market participants operating in the “shadows of our markets” (Kanjorski 2009). Private fund managers with assets under management (AUM) in excess of \$150 million were required to register with the SEC as investment advisers and had to disclose information about their trades and portfolios to the SEC (Dodd Frank Act §§ 408, 403, IA Release 3221, IA Release 3222). Required disclosures under Title IV include: counterparty credit risk exposures, trading and investment positions, trading practices, the amount of AUM, valuation policies, side letters, the use of leverage, and other information deemed necessary and appropriate to avoid systemic risk (Dodd-Frank Act §§ 404, 405).

2.1. Registration

Title IV of the Dodd-Frank Act exempts private fund advisers with less than \$150 million AUM from registration (Dodd-Frank Act § 408) and requires the SEC to examine factors including the investment strategy, size, and governance of an investment adviser in order to determine the systemic risk of private funds and to impose registration and examination procedures accordingly. To prevent the exemptions from registration to “swallow the rules,” Title IV authorizes the SEC to utilize its rulemaking authority (Kanjorski 2010). The SEC can require the disclosure of any other reports it considers necessary to protect investors (Dodd-Frank Act § 408).

Investment advisers registering with the SEC are required to file the pertinent disclosure document - Form ADV (Form ADV, Sorady 2011, Coakley and Allen 2011, Koehler and Lambert 2011). To implement the new registration requirements under Title IV, the SEC amended Form ADV. On March 30, 2012 investment advisers in the United States, including those who had previously registered with the SEC, had to file the SEC’s amendment to Form ADV (Rule 203A-5(b), IA Release 3221), reporting to the SEC information regarding the private funds they manage (Form ADV Part 1A). The SEC required previously exempt reporting advisers to file the necessary information between January 1 and March 30, 2012 (Rule 203A-5(b), IA Release 3221).

2.2. Disclosure

In addition to mandatory registration requirements, amended Form ADV requires disclosure of information regarding the fund structure, ownership, the gross asset value, the investment strategy, the scope of services provided, and the fund’s use of consultants and other gatekeepers (Amendments to Form ADV). Amended Form ADV also demands disclosure of non-advisory activities, financial industry affiliations, the number and types of its clients, and an assessment of the percentage of AUM attributable to each client type

(Form ADV Part 1A). In addition to the regular annual filing, investment advisers also have to update Form ADV if the disclosures therein become materially inaccurate (Form ADV Part 1A).¹

In addition to the filing requirements under Amended Form ADV, Title IV requires registered private fund advisers to file periodic reports (Dodd-Frank Act § 404, IA Release 3308). In January 2011, in a joint effort to implement the Dodd-Frank Act provisions, the SEC and the Commodity Futures Trading Commission (CFTC) together proposed Form PF (IA Release 3308). In October 2011, the SEC enacted Form PF (IA Release 3308, Rule 204(b)-1). In adopting the final rules, the SEC attempted to balance FSOC's interest in monitoring systemic risk through analyzing high quality information with industry concerns.

Form PF filing requirements apply to all registered investment advisers that manage RAUM in excess of \$150 million for private funds at the end of their most recently completed fiscal year (IA Release 3308). Private fund advisers managing less than \$1.5 billion RAUM attributable to private funds must file Form PF annually (IA Release 3308), whereas advisers managing RAUM in excess of \$1.5 billion attributable to private funds must file Form PF quarterly (IA Release 3308). The quarterly reporting requirement for large private fund advisers is intended to provide timely data that enables the FSOC to identify trends in systemic risk (IA Release 3308).

Mandatory Form PF disclosures include the following items: credit exposure, strategy, risks metrics, products used by the investment adviser, performance and changes in performance, financing information, the funds managed by the investment advisor, and information about individual investors and positions held by the investment advisor (Form PF). Counterparty credit exposure is an important item on Form PF, requiring private fund advisers to identify the five trading counterparties to which the reporting funds have the greatest net counterparty credit exposure (Form PF Section 1c, 2b).

3. Previous work on Private Fund Performance

Private fund performance has received a lot of attention in the literature. Prior studies have investigated the effects of various variables on private fund performance persistence. Several studies evaluate performance persistence in various contexts, (Fung et al. 2008, Jagannathan et al. 2010, Koh et al. 2003, Liang 2000 and 2003, Naik et al. 2007, Teo 2009, Agarwal and Naik 2000, Agarwal et al. 2006 and 2007, Baquero et al. 2005, Cassar and Gerakos 2009, Ding and Shawky 2007, Eling 2009, Fung and Hsieh 1997, 2000, 2001 and 2004), among others. Other studies consider fund regulation and governance in the United States (Hu and Black 2007, Kaal 2013a, 2015b, and forthcoming, Brown et al. 2008, Verret, 2008, Cassar and Gerakos, 2009, 2011), and internationally (Cumming and Dai 2009, 2010a and 2010b, Cumming and Johan 2008).

¹ Amended Form ADV obliges advisers to disclose advisory activities, clients, employees, compensation arrangements (IA Release 3221, Form ADV Part 1A). Once advisers file Form ADV they can no longer deduct accrued but unpaid liabilities and other outstanding debt from their totals, because rather than reporting net Regulatory Assets Under Management (RAUM), advisers are required under Amended Form PF to report their gross RAUM (IA Release 3221). Amended Form ADV also curtails investment advisers in their ability to exercise discretion in including or excluding assets from RAUM (Amended Form ADV, Part 1 A, instr. 5.b) and identifying the adviser's total RAUM and the ownership of RAUM by type of client (Form ADV Part 1A Item 5.D.(2)).

A significant part of the literature focuses on the performance of private funds in comparison with other market participants. Ackermann, McEnally and Ravenscraft (1999), for instance, find that while private funds are more volatile than mutual funds and market indices, they consistently outperform mutual funds, but not standard market indices. Incentive fees used by private funds may explain some of the higher performance, but cannot explain the increased total risk (Ackermann, McEnally and Ravenscraft 1999). Other studies evaluate the strategies employed by private fund managers and their impact on performance. De Los Rios, Diez and Garcia (2011), for instance, find that only a few private fund strategies provide significant value to investors and not all fund categories exhibit significant nonlinearities. Private fund managers' unique investment strategies are associated with better private fund performance (Sun, Wang and Zheng, 2012). Private fund strategy can also influence private funds' performance persistence (Elgin, 2009).

Absolute and relative performance of private funds is an important factor in the private fund industry. According to Amin and Kat (2003), private funds investments show the best results when 10%-20% of the portfolio value is invested in private funds but do not offer a superior risk-return profile as a stand-alone investment. Among more than three thousand private funds with similar style classification in 2011, Fung and Hsieh (2011) find that less than 20% of long/short equity private funds delivered persistent, significant, and stable positive non-factor related returns for investors. Past performance is associated with risk levels in private funds (Brown, Goetzmann and Park 2001). Absolute and relative performance of private funds is an important factor contributing to fund disappearance (Brown, Goetzmann and Park 2001). Average private fund returns are related positively to fund assets, lockup period, and incentive fees (Liang 1999). Funds performance and skills have been challenged by Lo and Hasanhodzic (2007) and Gramouridis and Paterlini (2010), that show that some private funds strategies can be replicated by combining simple financial tools.

Because private funds evolved in a regulatory environment with low or no regulatory supervision until the enactment of the Dodd-Frank Act, most prior studies on private fund performance do not assess the implications of private fund regulation. Prior studies have explored the impact of earlier attempts by the SEC to register private fund managers and increase disclosure requirements for private funds. In the context of the SEC's attempt in 2004 to register private fund advisers and file Form ADV disclosures (Kaal 2013a), Brown, Goetzmann, Liang and Schwarz (2008) investigate the effect of private fund manager registration on private funds' operational risk. The study concludes that market participants were already aware of operational risk. Cumming and Dai (2010) also analyze the impact of pre Dodd-Frank Act private fund regulation on fund structure and performance. They report that lower fund alphas, lower average monthly returns, and higher fixed fees are associated with the location of key service providers and permissible distributions.

The enactment of Dodd-Frank Act increases significantly regulatory oversight in private fund industry. Our study estimates the causal effect of an exogenous regulatory shock on private fund performance. More specifically, we estimate the causal effect of private fund manager registration and increased disclosure requirements under Title IV of the Dodd-Frank Act on private fund performance. Other studies investigate other effects of the Dodd-Frank Act. Among others, Dimmock and Gerken (2014) show that the

increased regulatory oversight introduced by the Dodd-Frank Act reduces returns misreport by private funds.²

4. DATA AND DESCRIPTIVE STATISTICS

To assess the impact of the registration effective date for private fund advisers under the Dodd-Frank Act on the private fund industry, we use data from the Morningstar Private Fund Database, Inc. on monthly private fund earnings (measured in US dollars) reported by about 7,000 private funds and more than 3,700 private fund advisers. Before the mandatory registration requirement for private fund advisers under the Dodd Frank Act, some private fund advisers voluntarily disclosed information about their investment strategies and earnings. After the mandatory registration requirement for private fund advisers under the Dodd Frank Act became effective on March 30, 2012, private fund advisers with AUM above \$150 million are required to comply with increased disclosure obligations. Private fund advisers with less than \$150 million AUM can choose to disclose information to the SEC.

Due to missing data and the presence of outliers, we extract a sample of 3,424 private funds that report all the monthly earnings data and monthly AUM in each period from January 2010 to September 2014. The data set contains information on individual private fund advisers and their managed funds (SEC identification number, name of the private fund, inception date, domicile) including monthly reported assets under management and monthly reported earnings.

The main descriptive statistics related to the monthly hedge funds returns and the logarithm of the AUM, reported in Table 1. We focus not only on the entire period but also on March 2012, the registration effective date for hedge fund advisers under the Dodd-Frank Act. The average number of funds with AUM larger than \$150 million constitutes 73% of the entire sample of funds. There are minor differences in the reported results between the entire period and March 2012. While the statistics related to the AUM are quite similar in the two considered periods, we notice that the mean performance in March 2012 is below the average of the entire sample period.

Because disclosure under the Dodd-Frank Act is mandatory only for the hedge fund advisers with an AUM exceeding the threshold set at \$150 million, we have divided our sample into two subsamples: the first subsamples includes all hedge funds with AUM always smaller or equal to \$150 million, the second subsample includes all fund with AUM larger than \$150 million. Such subsamples are referred to as "Small" and "Large" respectively.

INSERT TABLE 1

Figures 1 and 2 report the mean values for the hedge fund returns and the AUM (in logarithmic scale) for the two subsamples considered, while Table 2 and 3 display the main descriptive statistics for the entire period and for March 2012. Figure 1 shows that smaller funds outperform larger funds in the sample in eight of twelve months in 2012.

² The Dodd-Frank Act aims to increase stability of financial markets. See, for example, Kane (2012) and Balasubramnian and Cyree (2014) investigating the effect of Dodd-Frank Act on the stability of financial systems and market discipline on banks.

There is a strong variability in the performance of the subsamples, with on average no clear dominant group. Figure 2 shows that average AUM is stable across the entire period for both subsamples with slight variation for larger funds in the first quarter of 2012.

INSERT FIGURE 1

INSERT FIGURE 2

Table 2 provides more detailed information on the composition of the two subsamples in the sample period. About 73% of the entire sample of 3,424 hedge funds consists of funds with AUM larger than \$150 million. Only 26% represents the smaller subsample. While the mean return performances are similar for the two subsamples, the standard deviation and the minimum and maximum average performance suggest that the smaller subsample has a larger dispersion. When considering the AUM, both samples are homogeneous and quite stable.

INSERT TABLE 2

INSERT TABLE 3

The statistics for March 2012, shortly before the registration effective date for hedge fund advisers, March 30, 2012, show overall less dispersion. In March 2012, the larger fund subgroup reported a slightly lower standard deviation for Log (AUM) and Returns.

5.4. ESTIMATING THE EFFECT OF ADVISER REGISTRATION ON ADVISER PERFORMANCE

Our main identification strategy is based on the potential discontinuity generated by the enactment of the Dodd-Frank Act and SEC implementation rules that require private fund advisers with an AUM greater than \$150 million to comply with the registration requirement and file the necessary information between January 1 and March 30, 2012 (Rule 203A-5(b), IA Release 3221).

1. *Linear Regression*

As a first step, we estimate the parameters of a simple linear regression model, considering the entire sample of 3,424 funds for each period, from December 2011 to December 2012. We want to evaluate the possible presence of a relationship between the funds' returns (Y) and the logarithm of the AUM(X) at time t ($t = \text{Dec 2011}, \dots, \text{Dec 2012}$). As a second step, we introduce a dummy variable, which assumes value equal to 1 if the AUM is larger than \$150 million or 0 otherwise.

Tables 4 and 5 report the results of our investigation. We report the estimated values of the intercept (α), of the beta coefficient of the logarithm of the AUM (β) and of the dummy variable, when present. For each coefficient, we also report the corresponding p-values right below the coefficient. Finally, we report the value of the F-statistics with the corresponding p-value and the R-squared.

When considering the entire sample and no dummy variable, the estimated beta for the logarithm of the AUM are most statistically significant the period January – March 2012, e.g. right before the registration effective date for private fund advisers. We cannot reject the null hypothesis that the beta is equal to zero in the remaining period. The F-statistics also leads us to reject the null hypothesis that the coefficients are jointly equal to zero, supporting the validity of the model.

Examining the beta coefficients, we notice that in the period January - March 2012 and July 2012, beta coefficients are negative, while immediately after the registration effective date, e.g. April-May 2012 the beta coefficient is positive. In the period before the registration effective date for hedge fund advisers under the Dodd-Frank Act, e.g. March 2012, the size of funds seems to have a negative relationship with the fund performance, and a positive relationship thereafter. We conclude that size of the funds in our sample does not appear to matter for fund returns as only a few coefficients are statistically significant but are still close to zero.

INSERT TABLE 4

The simple linear regression models in Table 5 also include a dummy variable, with value equal to 1 for funds with AUM larger than \$150 million and 0 otherwise. Right before the registration effective date, in January and February 2012, the dummy variable results are not statistically significant and negative. The dummy variables are statistically significant and positive in April and September 2012. The F-statistics support the validity of the models, while the explanatory power, measured by the R-squared is still very limited. The Beta coefficients are all close to zero and at 5%. We reject the null hypothesis that the beta coefficients are statistically different from zero only for a few months.

The empirical results from the simple regression analysis, including the dummy analysis, seem to suggest that close to the registration effective date for hedge fund advisers under the Dodd-Frank Act, the size of AUM, above or below the regulatory threshold of \$150 million, does not play a significant role in explaining hedge fund returns of the entire sample.

INSERT TABLE 5

2. *Regression Discontinuity*

Our empirical methodology is based on a regression discontinuity design, using the Rubin Causal Model (RCM) (Rubin 1974, Imbens and Rubin 2007, Imbens and Wooldridge 2007). The regression discontinuity design allows us to examine the causal effect of a binary treatment of units (which may be individuals, firms, or other entities), that may have been exposed or may not have been exposed to a specific treatment.³ In our analysis, the unit of observation is a specific private fund before and after the treatment, represented by the registration effective date for private fund advisers under the Dodd-Frank Act, March 30, 2012. Additionally, the treatment may affect the units, each private fund adviser in our sample, in a heterogeneous way, which needs to be controlled for.

³ The RD methodology has been applied widely in the law and economics literature in order to identify the effect on an exogenous change in legislation. See Carpenter and Dobkin (2009), Black et al. (2012) and MacDonald et al. (2015) among others.

In the following, we use the Sharp Regression Discontinuity (SRD) design (Trochim 1984 and 2001, Imbens and Kalyanaraman 2011). Our analysis has been performed in STATA, using the code described in Fuji, Imbens and Kalyanaraman (2009) and the STATA packages **rd** (Nichols 2007) and **rdrobust** (Calonico et al. 2013).

In the SRD design, the assignment of units to the treatment group is governed by the variable W_i , defined as a deterministic function of the forcing (or treatment-determining) variable X and the fixed threshold c , and takes the following value:

$$W_i = \begin{cases} 1 & \text{if } X_i > c \\ 0 & \text{otherwise} \end{cases}$$

All units with a value of X_i at least equal to c are assigned to the treatment group. On the contrary, all units with a covariate value less than c are assigned to the control group (and are not eligible for the treatment). The basic idea behind the RD design is that any discontinuity in the conditional distribution of Y_i as a function of X_i at the cutoff value c is interpreted as evidence of a causal effect of the treatment.

In the context described here, the registration of the private fund advisers under the Dodd-Frank Act is the treatment. The Dodd-Frank Act requires a private fund adviser to be registered if the AUM exceeds \$150 million after March 30, 2012, the registration effective date. Private fund advisers are assigned to the treatment group only when these two conditions are met. The outcome variable Y_i is the private fund advisers' returns in each period of the sample, after the registration effective date for private fund advisers under the Dodd-Frank Act. The forcing variable X_i is the log AUM of each private fund adviser i , with the threshold c set equal to \$150 million, which has been re-scaled to zero in the following figures.

We apply the data-driven optimal choice of evenly-spaced bins in Calonico et al. (2013). The procedure aims to use bins to approximate the underlying regression functions by local sample means. The optimal solutions provided by their approach would suggest the partition of our data into seven bins, which is too small. Hence, we increased the optimal number of bins by a factor of 20 and 100 to obtain a plot showing a cloud of points as shown in Figure 5 below. We notice that the variability on the returns (y-axis) is larger for the funds with large AUM and as we increase the number of bins, the regression function clearly gets smoother. However, the presence of discontinuity is evident, no matter what number of bins we consider.

Under a SRD approach, following Imbens and Kalyanaraman (2012), we focus on the estimation of the average effect of the treatment for units with covariate values equal to the threshold, denoted with coefficient τ_{RD} . We perform the regression on the entire sample from January 2010 to August 2014.

We have extended our analysis using a Fuzzy Regression Discontinuity (FRD) approach on the entire sample. FRD design allows us to model the probability of receiving the treatment (compliance with mandatory information disclosure) as a smaller jump at the threshold (i.e. lower than 1), rather than a sharp change from 0 to 1 as in the SRD design. FRD allows us to check whether the discontinuity has occurred in a period different from March 30, 2012. The reason for implementing FRD is that hedge fund advisers may have anticipated the costs of compliance with mandatory disclosure in the months preceding the enactment of the Dodd Frank Act. However, because the

denominator is very close to 1 in our investigation, empirical results show that the differences between the SRD and the FRD approach are of minor importance. Our estimates are computed in correspondence of the optimal bandwidth. As a robustness check, we computed such estimates when considering different bandwidths.

Using an array of robustness test validating our RD results, Figures 3-7 suggest that the requirements introduced by the Dodd-Frank Act create no significant effect on private fund performance. The P-values for all RD results in Table 6 are above the 5% level and confirm our finding of no effect. By contrast, the private fund industry expected the introduction of the Dodd-Frank Act to result in negative effects on private fund returns. The absence of any statistical significant effect of mandatory disclosure on hedge fund returns may suggest that the transparency costs associated with disclosure do not significantly affect the profitability of hedge fund advisers.

INSERT FIGURE 3

INSERT FIGURE 4

INSERT TABLE 6

INSERT FIGURE 5

INSERT FIGURE 6

INSERT FIGURE 7

5.5. DISCUSSION & FUTURE RESEARCH

For much of its history, the private fund industry has viewed private fund adviser registration and the disclosure of proprietary information as a threat to its profitability. Regulators have attempted for decades to increase the monitoring and supervision of the private fund industry. Ending the struggle between the industry and regulators, Title IV of the Dodd-Frank Act introduced a mandatory registration and disclosure requirement for private fund managers.

Prior studies have demonstrated that mandatory private fund adviser registration under the Dodd-Frank Act affects the cost structure of the industry (Kaal 2013a, 2015b, 2015c, 2016) collected data on private fund managers, showing that registration and increased compliance requirements under the Dodd-Frank Act marginally increase the cost structure of private funds. Kaal (2013a, 2015c) finds non-robust evidence that the higher administrative costs imposed by the Dodd-Frank Act are a second-order effect of the regulation, thereby not affecting the overall returns of private funds. We are not aware of any other empirical evidence on the effects of the Dodd-Frank Act on the private fund industry. Compared with the existing prior work, in this study, we use a much larger dataset and a more sophisticated empirical approach such as regression discontinuity. We find no statistical evidence for an effect of the requirements introduced by the Dodd-Frank Act on private fund advisers' performance.

Acknowledgements

The authors would like to thank Bernard Black, Katherine Litvak, Douglas Cumming, Shari Seidman Diamond, Ronald Masulis, CNV Krishnan, Scott Hirst, Dhammika Dharmapala, David Zaring, John Morley, Dirk Zetsche, Tamar Frankel, William Bratton, Sabastian V. Niles, Anne Tucker, Joan Heminway, Jennifer Taub, James Cox, and Lisa Fairfax, the participants at the 2013 Conference on Empirical Legal Studies, the participants at the 2013 annual meeting of the Midwest Law & Economics Association, the participants at the Fourth Annual Junior Faculty Business and Financial Law Workshop at George Washington University Law School, the participants at the 2014 annual meeting of the Eastern Finance Association, the participants at the 2014 annual meeting of the European Financial Management Association, and the participants at the 2015 and 2016 annual meeting of the American Law & Economics Association.

References

- 155 Cong. Rec. H14419-20 (daily ed. Dec. 9, 2009) (statement of Rep. Paul Kanjorski).
- 155 Cong. Rec. H14441-42 (daily ed. Dec. 9, 2009) (statement of Rep. Sheila Jackson Lee).
- 156 Cong. Rec. H5233, H5235-39 (daily ed. June 30, 2010) (statement of Rep. Paul Kanjorski).
- 156 Cong. Rec. S5902, S5925 (daily ed. July 15, 2010) (statement of Sen. Dick Durbin).
- 17 C.F.R. § 275.203A-5(b) (2014).
- 17 C.F.R. § 279.9 (2014).
- 17 C.F.R. § 275.203A-5 (2014).
- Ackermann, Carl, Richard McEnally, and David Ravenscraft. 1999. "The Performance of Hedge Funds: Risk, Returns, and Incentives." *Journal of Finance* 54:833-874. doi:10.1111/0022-1082.00129.
- Agarwal, Vikas, and Narayan Y. Naik. 2000. "Multi-Period Performance Persistence Analysis of Hedge Funds." *Journal of Financial and Quantitative Analysis* 35:327-342. doi:10.2307/2676207.
- Agarwal, Vikas, and Narayan Y. Naik. 2006. "Flows, Performance, and Managerial Incentives in the Hedge Fund Industry." London Business School Working Paper.
- Agarwal, Vikas, and Narayan Y. Naik. 2007. "Why is Santa So Kind to Hedge Funds? The December Return Puzzle." London Business School Working Paper.
- Agarwal, Vikas, Naveen D. Daniel, and Narayan Y. Naik. 2009. "Role of Managerial Incentives and Discretion in Hedge Fund Performance." *Journal of Finance* 64: 2221-2256. doi:10.1111/j.1540-6261.2009.01499.
- Amenc, Noël, and Lionel Martellini. 2003. "The Alpha and Omega of Hedge Fund Performance Measurement." Risk and Asset Management Research Center, EDHEC Business School, Lille, France. <http://goo.gl/77VdtK>.
- Amenc, Noël, Sina El Bied, and Lionel Martellini. 2003. "Predictability in Hedge Fund Returns." *Financial Analysts Journal* 59: 32-46.
- Amendments to Form ADV, Investment Advisers Act Release No. 3060, 75 Fed. Reg. 49,234 (July 28, 2010) (IA Release 3060).
- Americas' Financial Services Regulatory Center of Excellence (CoE). 2011. "Regulatory Reform for the Financial Services Industry--and Beyond." KPMG, issued May 4, 2011, and accessed December 3, 2012. <http://goo.gl/77VdtK>.
- Amin, Gaurav S., and Harry M. Kat. 2003. "Hedge Fund Performance 1990-2000: Do the 'Money Machines' Really Add Value?" *Journal of Financial and Quantitative Analysis* 38:251-274. doi:10.2307/4126750.
- Balasubramnian, Bhanu, and Ken B. Cyree. 2014. "Has market discipline on banks improved after the Dodd-Frank Act?" *Journal of Banking and Finance* 41:155-166.
- Baquero, Guillermo, Jenke ter Horst, and Marno Verbeek. 2005. "Survival, Look-Ahead Bias, and Persistence in Hedge Fund Performance." *Journal of Financial and Quantitative Analysis* 40:493-517.
- Black, Bernard, and Woonchan Kimb. 2012. "The effect of board structure on firm value: a multiple identification strategies approach using Korean data." *Journal of Financial Economics* 104:203-226.

- Brown, Stephen J., William N. Goetzmann, and James Park. 2001. "Careers and Survival: Competition and Risk in the Hedge Fund and CTA Industry." *Journal of Finance* 56:1869-1886. doi:10.1111/0022-1082.00392.
- Brown, Stephen, William Goetzmann, Bing Liang, and Christopher Schwarz. 2008. "Mandatory Disclosure and Operational Risk: Evidence from Hedge Fund Registration." *Journal of Finance* 63:2785-2815. doi: 10.1111/j.1540-6261.2008.01413.x.
- Carpenter, Christopher, and Carlos Dobkin. 2009. "The Effect of Alcohol Consumption on Mortality: Regression Discontinuity Evidence from the Minimum Drinking Age." *American Economic Journal: Applied Economics* 1:164-182.
- Cassar, Gavin, and Joseph Gerakos. 2009. "Determinants of Hedge Fund Internal Controls and Fees." *The Accounting Review* 85:1887-1919.
- Cassar, Gavin, and Joseph Gerakos. 2011. "Hedge Funds: Pricing Controls and the Smoothing of Self-Reported Returns." *Review of Financial Studies* 24:1698-1734.
- Coakley, Michael P., and Matthew P. Allen. 2011. "The New Form ADV Part 2 and the 'Plain English' Movement of the SEC, FINRA and Michigan's OFIR." *Michigan Business Law Journal* 31:19-30.
- Calonico, Sebastian, Matias D. Cattaneo, and Rocio Titiunik. 2013. "Robust Data-Driven Inference in the Regression-Discontinuity Design." *The Stata Journal* vv(ii):1-34.
- _____. 2014. "Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs." *Econometrica* 82:2295-2326.
- Commission of the European Communities. 2009. "Commission Proposal for a Directive of the European Parliament and of the Council on Alternative Investment Fund Managers and amending Directives 2004/39/EC and 2009/.../EC." <http://goo.gl/beP3ND>.
- Cumming, Douglas, and Na Dai. 2009. "Capital Flows and Hedge Fund Regulation." *Journal of Empirical Legal Studies* 6:848-873.
- _____. 2010a. "A Law and Finance Analysis of Hedge Funds." *Financial Management* 39:997-1026. doi:10.1111/j.1755-053X.2010.01102.x.
- _____. 2010b. "Hedge Fund Regulation and Misreported Returns." *European Financial Management* 16:829-867.
- Cumming, Douglas, and Sophia A. Johan. 2008. "Hedge Fund Forum Shopping." *University of Pennsylvania Journal of Business Law* 10:783-831.
- De Los Rios, Antonio Diez, and René Garcia. 2011. "Assessing and Valuing the Nonlinear Structure of Hedge Fund Returns." *Journal of Applied Econometrics* 26:193-212. doi:10.1002/jae.1147.
- Department of the Treasury. 2009. *Financial Regulatory Reform: A New Foundation: Rebuilding Financial Supervision and Regulation*. <http://goo.gl/JUHEb>.
- Dimmock, Stephen G., and William C. Gerken. 2014. "Regulatory Oversight and Return Misreporting by Hedge Funds.", mimeo.
- Ding, Bill, and Hany A. Shawky. 2007. "The Performance of Hedge Fund Strategies and the Asymmetry of Return Distributions." *European Financial Management* 13:309-331.
- Dodd-Frank Wall Street Reform and Consumer Protection Act, Pub. L. No. 111-203, §§ 401, 402, 408, 124 Stat. 1326 (2010).

- Economist*. 2010. "A Few Minutes of Mayhem: America's Stockmarket Plunge." May 15, United States edition, 82.
- Eling, Martin. 2009. "Does Hedge Fund Performance Persist? Overview and New Empirical Evidence." *European Financial Management* 15:362-401. doi:10.1111/j.1468-036X.2008.00471.x.
- Ernst and Young. 2012. "Global Hedge Fund and Investor Survey 2012." Accessed November 28. <http://goo.gl/DGY61h>.
- Exemptions for Advisers to Venture Capital Funds, Hedge Fund Advisers with Less Than \$150 Million in Assets Under Management, and Foreign Hedge Advisers, Investment Advisers Act Release No. 3222, 76 Fed. Reg. 39,646 (July 6, 2011) (IA Release 3222).
- Ferullo, Mike, R. Christian Bruce, Richard Hill, and Malini Manickavasgam. 2010. "Senate Sends Regulatory Reform Bill To White House for President's Signature." *Securities Regulation and Law Report* 42:1353.
- French, Kenneth R. 2008. "The Cost of Active Investing." *Journal of Finance* 63:1537–1573.
- Fuji, Daisuke, Guido Imbens and Karthik Kalyanaraman. 2009. Notes for MATLAB and STATA Regression Discontinuity Software, Technical Report.
- Fung, William, and David A. Hsieh. 1997. "Empirical Characteristics of Dynamic Trading Strategies: The Case of Hedge Funds." *Review of Financial Studies* 10:275-302.
- _____. 2000. "Performance Characteristics of Hedge Funds and CTA Funds: Natural Versus Spurious Biases." *Journal of Financial and Quantitative Analysis* 35:291-307.
- _____. 2001. "The Risk in Hedge Fund Strategies: Theory and Evidence from Trend Followers." *Review of Financial Studies* 14:313-341.
- _____. 2002. "Risk in Fixed-Income Hedge Fund Styles." *Journal of Fixed Income* 12:6-27. doi:10.3905/jfi.2002.319321.
- _____. 2004. "Hedge Fund Benchmarks: A Risk Based Approach." *Financial Analysts Journal* 60:65–80.
- _____. 2011. "The Risk in Hedge Fund Strategies: Theory and Evidence from Long/Short Equity Hedge Funds." *Journal of Empirical Finance* 18:547-569. doi:10.1016/j.jempfin.2011.04.001.
- Fung, William, David A. Hsieh, and Tarun Ramadorai. 2008. "Hedge Funds: Performance, Risk, and Capital Formation." *Journal of Finance* 63:1777-1803.
- Goldfarb, Zachary A. and David Cho. 2009. "Hedge Funds Making Way for Government Regulation; Industry Faces New Realities, Investors Who Want Assurances." *Washington Post*, March 14, D.2.
- Hu, T.C. Hu, and Bernard Black. 2007. "Hedge Funds, Insiders, and the Decoupling of Economic and Voting Ownership: Empty Voting and Hidden (Morphable) Ownership." *Journal of Corporate Finance* 13:343-367.
- Imbens, Guido and Thomas Lemieux. 2008. "Regression discontinuity designs: A guide to practice." *Journal of Econometrics*. 142:615–635.
- Imbens, Guido W., and Karthik Kalyanaraman. 2011. "Optimal Bandwidth Choice for the Regression Discontinuity Estimator." *Review of Economic Studies* 79:933-959.
- Investment Advisers Act of 1940, Pub. L. No. 76-768, 54 Stat. 847 (1940).
- Investment Company Act of 1940, Pub. L. No. 76-768, 54 Stat. 789 (1940).

- Jagannathan, Ravi, Alexey Malakhov, and Dmitry Novikov. 2010. "Do Hot Hands Exist Among Hedge Fund Managers? An Empirical Evaluation." *Journal of Finance* 65:217–255. doi:10.1111/j.1540-6261.2009.01528.x.
- Jickling, Mark, and Alison A. Raab. 2006. *Hedge Fund Failures*. Congressional Research Service, RL33746.
- Jones, Alfred Winslow. 1949. "Fashions in Forecasting." *Fortune*, March, 88.
- Jorion, Philippe. 2000. "Risk Management Lessons from Long-Term Capital Management." *European Financial Management* 6:277-300. doi:10.1111/1468-036X.00125.
- Kaal, Wulf A. 2009. "Hedge Fund Valuation: Retailization, Regulation, and Investor Suitability." *Review of Banking and Financial Law* 28:581-637.
- _____. 2011. "Hedge Fund Regulation Via Basel III." *Vanderbilt Journal of Transnational Law* 44:389-463.
- _____. 2013a. "Hedge Fund Manager Registration Under the Dodd-Frank Act." *San Diego Law Review* 50:243-318.
- _____. 2015b. "Hedge Fund Disclosures under the Dodd-Frank Act." *Brooklyn Journal of Corporate, Financial & Commercial Law* 9:428-477.
- _____. 2015c. "What Drives Dodd-Frank Act Compliance Cost for Private Funds?" U of St. Thomas (Minnesota) Legal Studies Research Paper No. 15-13. *Journal of Alternative Investments*, forthcoming.
- _____. 2016. "The Post Dodd-Frank Act Evolution of the Private Fund Industry – Comparative Evidence from 2012 and 2015." U. of St. Thomas (Minnesota) Legal Studies Research Paper.
- Kane, Edward J. 2012. "Missing elements in US financial reform: A Kübler-Ross interpretation of the inadequacy of the Dodd-Frank Act." *Journal of Banking and Finance* 36:654-661.
- Kaulessar, Ricardo. 2012. "Survey: Fund Managers Welcome Regulations – With Reservations." eVestment Alliance, last modified April 13, and accessed December 4. <http://goo.gl/Bd4O57>.
- Koehler, James F., and P. Wesley Lambert. 2011. "Impact of the Dodd-Frank and Registration Acts of 2010 on Investment Advisers." *Duquesne Business Law Journal* 13:29-42.
- Koh, Francis, Winston T.H. Koh, and Melvyn Teo. 2003. "Asian Hedge Funds: Return Persistence, Style and Fund Characteristics." Working paper. Singapore Management University.
- Landau, Peter. 1968. "Alfred Winslow Jones: The Long and the Short of the Founding Father." *Institutional Investor*, August.
- Lee, David S., and Thomas Lemieux. 2009. "Regression Discontinuity Designs in Economics." Working Paper. Department of Economics, Princeton University.
- Liang, Bing. 1999. "On the Performance of Hedge Funds." *Financial Analysts Journal* 55:72-85.
- _____. 2000. "Hedge Funds: The Living and the Dead." *Journal of Financial and Quantitative Analysis* 35:309-326.
- _____. 2003. "The Accuracy of Hedge Fund Returns." *Journal of Portfolio Management* 29:111-122.

- Lo, Andrew and Jasmina Hasanhodzic. 2007. "Can Hedge-Fund Returns Be Replicated?: The Linear Case." *Journal of Investment Management* 5:5–45.
- Loomis, Carol J. 1966. "The Jones Nobody Keeps Up With." *Fortune*, April.
- Nomination of William H. Donaldson: Hearing Before the Comm. on Banking, Housing and Urban Affairs, 108th Cong. 1, 37 (2003).
- MacDonald, John M., Jonathan Klick and Ben Grunwald. 2015. "The Effect of Privately Provided Police Services on Crime: Evidence from a Geographic Regression Discontinuity Design." *Journal of the Royal Statistical Society: Series A*.
- McCrary, Justin. 2008. "Manipulation of the running variable in the regression discontinuity design: A density test." *Journal of Econometrics* 142:698-714.
- Naik, Narayan Y., Tarun Ramadoria and Maria Stromqvist. 2007. "Capacity Constraints and Hedge Fund Strategy Returns". *European Financial Management* 13:239-256.
- Nichols, Austin. 2007. "Causal Inference with Observational Data." *The Stata Journal*, 7:507-541.
- Oesterle, Dale A. 2006. "Regulating Hedge Funds." *Entrepreneurial Business Law Journal* 1:1-42.
- Paletta, Damian, and Aaron Lucchetti. 2010. "Law Remakes U.S. Financial Landscape." *Wall Street Journal*, July 16, New York edition, A.1.
- Philips, Christopher B., Francis M. Kinniry Jr., David J. Walker, Todd Schlanger, Joshua M. Hirt. 2014. The Case for Index-Fund Investing. *Vanguard Research*. <https://goo.gl/k7KGnF>.
- Prohibition of Fraud by Advisors to Certain Pooled Investment Vehicles; Accredited Investors in Certain Private Investment Vehicles, Investment Advisers Act Release No. 2576, 72 Fed. Reg. 400 (January 4, 2007) (IA Release 2576).
- Prohibition of Fraud by Advisors to Certain Pooled Investment Vehicles, Investment Advisers Act Release No. 2628, 72 Fed. Reg. 44,756 (August 9, 2007) (IA Release 2628).
- Registration Under the Advisers Act of Certain Hedge Fund Advisers, Investment Advisers Act Release No. 2333, 69 Fed. Reg. 72,054, 72,059 (December 10, 2004) (IA Release 2333).
- Regulating Hedge Funds and Other Private Investment Pools: Hearing Before the United States Subcomm. on Securities, Insurance, and Investment, 111th Cong. 1 (2009) (statement of Jack Reed, Chairman, Senate Comm. on Banking, Housing, and Urban Affairs, Subcomm. on Securities, Insurance, and Investment).
- Regulation of Hedge Funds: Hearings Before the United States Senate Comm. on Banking, Housing, and Urban Affairs, 109th Cong. 7 (2006) (statement of Christopher Cox, Chairman, Securities and Exchange Commission).
- Reporting by Investment Advisers to Private Funds and Certain Commodity Pool Operators and Commodity Trading Advisors [sic] on Form PF, Investment Advisers Act Release No. 3308 (October 31, 2011), 76 Fed. Reg. 71,128 (November 16, 2011) ([IA Release 3308](#)).
- Reporting Form for Investment Advisers to Private Funds and Certain Commodity Pool Operators and Commodity Trading Advisors. <http://www.sec.gov/about/forms/formpf.pdf>. Accessed December 4, 2012 (Form PF).

- Roth, Paul N., and Brian H. Fortune. 2001. "Hedge Fund Regulation in the Aftermath of Long Term Capital Management." in *Hedge Funds: Law and Regulation*, edited by Iain Cullen and Helen Parry. London: Sweet and Maxwell.
- Rules Implementing Amendments to the Investment Advisers Act of 1940, Investment Advisers Act Release No. 3221, 76 Fed. Reg. 42,950 (July 19, 2011) (IA Release 3221).
- Securities and Exchange Commission. 2003. *Implications of the Growth of Hedge Funds*. <http://goo.gl/T3vI6m>.
- Uniform Application for Investment Adviser Registration and Report by Exempt Reporting Advisers. Accessed December 4, 2012. <http://www.sec.gov/about/forms/formadv.pdf>. (Form ADV).
- Sorady, Marybeth, Daren Domina, Wendy Cohen, Fred Santo, Henry Bregstein, Meryl Wiener, Marilyn Okoshi, and Jack P. Governale. 2011. "Summary and Analysis of Dodd-Frank Rules for Investment Advisers: Registration Requirements, Exemptions, Family Offices, Performance Fee Eligibility." *Journal of Investment Compliance* 12:4-17. doi:10.1108/15285811111188126.
- Strasburg, Jenny. 2009. "U.S. would have Hedge Funds Register and Disclose Clients." *Wall Street Journal*, February 3, Asia edition, M.3.
- Sun, Zheng, Ashley Wang, and Lu Zheng. 2012. "The Road Less Traveled: Strategy Distinctiveness and Hedge Fund Performance." *Review of Financial Studies* 25: 96-143. doi:10.1093/rfs/hhr092.
- Teo, Melvyn. 2009. "The Geography of Hedge Funds." *Review of Financial Studies* 22:3531-3561.
- Trochim, William M.K. 1984. *Research Design for Program Evaluation: The Regression-Discontinuity Design*. Sage Publications: Beverly Hills, CA.
- Trochim, William M.K. 2001. "Regression-discontinuity design." In *International Encyclopedia of the Social and Behavioral Sciences*, vol. 19, 12940-12945, edited by Neil J. Smelser and Paul B. Baltes. New York: Elsevier.
- Van Der Klaauw, Wilbert. 2008. "Regression-Discontinuity Analysis: A Survey of Recent Developments in Economics," *Labour*, 22:219-245.
- Verret, J.W. 2007. "Dr. Jones and the Raiders of Lost Capital: Hedge Fund Regulation, Part II, A Self-Regulation Proposal." *Delaware Journal of Corporate Law* 32: 799-841.

APPENDIX

TABLES

TABLE 1

Variable	Average Value	March 2012
Number of funds	920	920
≤150 mil		
Funds with	2502	2502
AUM>150 mil		
RETURNS		
Min	-26.12	-15.99
Mean	0.68	0.57
Max	55.34	33.77
Std	3.94	3.45
LOG (AUM)		
Min	9.21	9.21
Mean	17.07	17.08
Max	23.87	23.84
Std	1.90	1.89

Table 1 – *Descriptive Statistics on the Entire Sample, January-December 2012*

TABLE 2

Variable	Fund AUM ≤150 mil	Fund AUM >150 mil
Number of Funds	920	2502
RETURNS		
Min	-26.12	-13.78
Mean	0.71	0.64
Max	54.26	24.16
Std	4.35	3.09
LOG (AUM)		
Min	9.21	11.71
Mean	16.56	19.15
Max	19.46	23.87
Std	1.56	1.67

Table 2 – Descriptive Statistics Average Values for Large and Small Subsamples, January-December 2012

TABLE 3

Variable	Fund AUM ≤150 mil	Fund AUM >150 mil
Number of Funds	920	2502
RETURNS		
Min	-15.99	-12.18
Mean	0.54	0.61
Max	27.21	33.77
Std	3.55	3.27
LOG (AUM)		
Min	9.21	18.83
Mean	16.54	19.73
Max	18.82	23.84
Std	1.56	0.77

Table 3 – Descriptive Statistics Small and Large Subsamples, March 2012

TABLE 4

Regression Statistics Linear Model (Dec 2011 to Dec 2012)													
	Dec 2011	Jan 2012	Feb 2012	Mar 2012	Apr 2012	May 2012	Jun 2012	Jul 2012	Aug 2012	Sept 2012	Oct 2012	Nov 2012	Dec 2012
Alpha	-0.2157185 (0.832)	9.857041*** (0)	2.779961*** (0.001)	2.117204** (0.024)	-1.092872 (0.153)	-5.167892*** (0.002)	2.073734* (0.057)	1.318737 (0.468)	3.762679*** (0.00)	0.339569 (0.694)	-2.184016*** (0.014)	-1.434683 (0.14)	1.504264 (0.157)
Beta	0.00324 (0.956)	-0.4049383*** (0)	-0.0489703 (0.305)	-0.093604* (0.085)	0.0454055 (0.307)	0.1906136** (0.046)	-0.0929947* (0.141)	-0.0152903 (0.885)	-0.1432312*** (0.004)	0.0507516 (0.312)	0.0988003* (0.057)	0.0905855 (0.109)	-0.0080059 (0.897)
F-Stat	0.00 (0.9563)	13.36*** (0.0003)	1.05 (0.3046)	2.97* (0.0851)	1.04 (0.307)	4** (0.0457)	2.17 (0.1414)	0.02 (0.885)	8.28*** 0.0041	1.02 (0.3117)	3.64* (0.0567)	2.58 (0.1088)	0.02 (0.8965)
R-Squared	0.00	0.012	0.001	0.0027	0.001	0.0037	0.002	0.00	0.0078	0.001	0.0036	0.0026	0.00

Table 4 - Regression statistics for a linear model in the periods December 2011 through December 2012. P-values are reported in parentheses. *, **, and *** indicate significance at 10%, 5%, and 1% levels. Significant results (at 10% level or better) are shown in boldface.

TABLE 5

Regression Statistics Linear Model with Dummy (Dec 2011 to Dec 2012)

	Dec 2011	Jan 2012	Feb 2012	Mar 2012	Apr 2012	May 2012	Jun 2012	Jul 2012	Aug 2012	Sept 2012	Oct 2012	Nov 2012	Dec 2012
Alpha	0.0614465 (0.957)	9.618512*** (0.000)	2.391477** (0.015)	2.470767** (0.035)	0.1809835 (0.845)	-5.616524*** (0.004)	0.6660763 (0.594)	1.558515 (0.452)	3.871995*** (0.00)	-0.6925546 (0.475)	-2.064664** (0.042)	-1.541117 (0.159)	0.8438437 (0.48)
Beta	-0.0151869 (0.825)	-.3894468*** (0.004)	-0.0239626 (0.684)	-0.1160514* (0.098)	-0.0361457 (0.515)	0.2196748* (0.059)	-0.001148 (0.988)	-0.0310213 (0.803)	-0.1504024*** (0.01)	0.1188031** (0.041)	0.0909656 (0.134)	0.0976125 (0.136)	0.0355217 (0.618)
Dummy	0.1657301 (0.599)	-.1299044 (0.841)	-0.2068031 (0.47)	0.1779477 (0.613)	0.6696762** (0.015)	-0.2454423 (0.661)	-0.7997703** (0.024)	0.1412264 (0.809)	0.0649386 (0.813)	-0.6365251** (0.021)	0.0715396 (0.803)	-0.0650892 (0.832)	-0.3938859 (0.227)
F-Stat	0.14 (0.8697)	6.70*** (0.0013)	0.79 (0.4548)	1.61 (0.1999)	3.49** (0.0308)	2.1 (0.1235)	3.65** (0.0264)	0.04 (0.9611)	4.16** (0.0158)	3.19** (0.0417)	1.85 (0.1579)	1.31 (0.2704)	0.74 (0.4784)
R-Squared	0.0003	0.0120	0.0014	0.0029	0.0064	0.0038	0.0067	0.0001	0.0079	0.0061	0.0036	0.0026	0.0015

Table 5 - Regression statistics for a linear model with dummy variable in the periods December 2011 through December 2012. P-values are reported in parentheses. *, **, and *** indicate significance at 10%, 5%, and 1% levels. Significant results (at 10% level or better) are shown in boldface.

TABLE 6

	Measure	Coef.	P-Value
Entire Sample	Conventional RD	0.95106	0.118
	Bias-corrected RD	1.1581	0.057
	Robust RD	1.1581	0.098
Mar-12	Conventional RD	0.95106	0.118
	Robust RD	N/A	0.098
	Robust with Cross Validation	N/A	0.104
	Optimal Bandwidth 1.2006146	0.8884178	0.107
	Bandwidth 0.30015365	0.9475949	0.363
	Bandwidth 0.60030731	0.4632891	0.566
	Bandwidth 0.90046096	0.8731405	0.178
	Bandwidth 1.5007683	0.6539375	0.184
	Bandwidth 1.8009219	0.4522598	0.319
	Bandwidth 2.1010756	0.2725607	0.521
	Bandwidth 2.4012292	0.1864635	0.644
	Optimal BW Feb. 2012 Return as Data	0.0565339	0.915
	Optimal BW Jan. 2012 Return as Data	0.2624185	0.733
	Optimal BW Dec. 2011 Return as Data	0.2087388	0.656
	Optimal BW Feb. 2012 Return as X	0.0565339	0.915
	Optimal BW Jan. 2012 Return as X	0.2624185	0.733
	Optimal BW Dec. 2011 Return as X	0.2087388	0.656
	Optimal BW Apr. 2012 Return as X	0.6087372	0.221
	Optimal BW May 2012 Return as X	-2.655924	0.039
	Optimal BW Jun. 2012 Return as X	0.1311312	0.861

Table 6 - Regression Discontinuity Coefficients and P-Values for entire sample and in March 2012 with bandwidth (BW) changes. *, **, and * significance at 10%, 5%, and 1% levels. Significant results (at 10% level or better) are shown in **boldface**.**

FIGURES

FIGURE 1

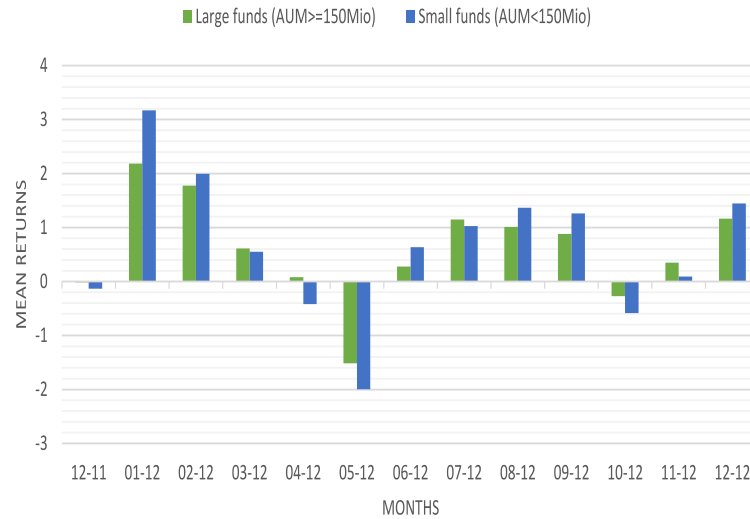


Figure 1 – Average Mean Hedge Fund Returns, January-December 2012

FIGURE 2

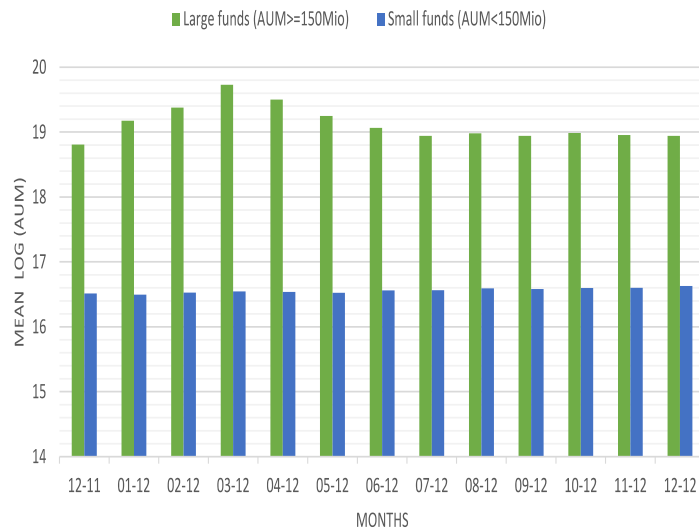


Figure 2 – Mean Hedge Fund Logarithm AUM, January-December 2012

FIGURE 3

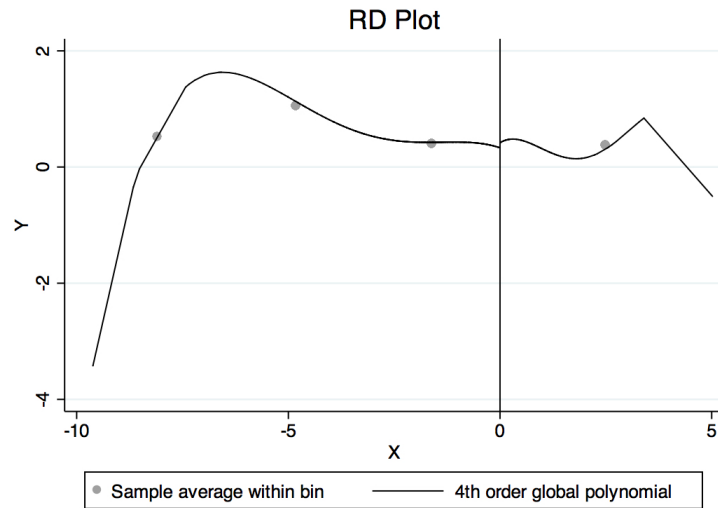


Figure 3 –RD plot using scaled down optimal bin-length choice relative to the entire sample of [2145] funds in March 2012. X-axis: logarithm of the (AUM) after subtracting the threshold of $\log(150\text{mil})$, y-axis: log-returns in March 2012

FIGURE 4

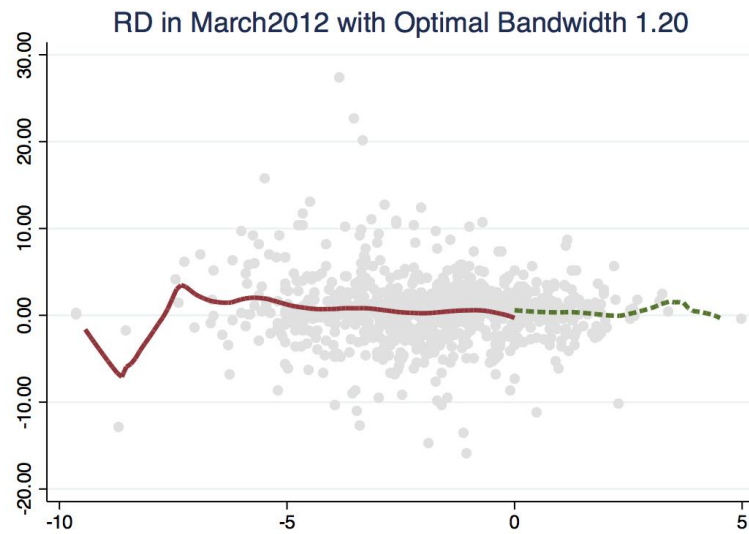


Figure 4 – SRD graph for registration effective date March 30, 2012. Figure 8 illustrates the discontinuity effect on private fund earnings that occurred on the registration effective date, March 30, 2012.

FIGURE 5

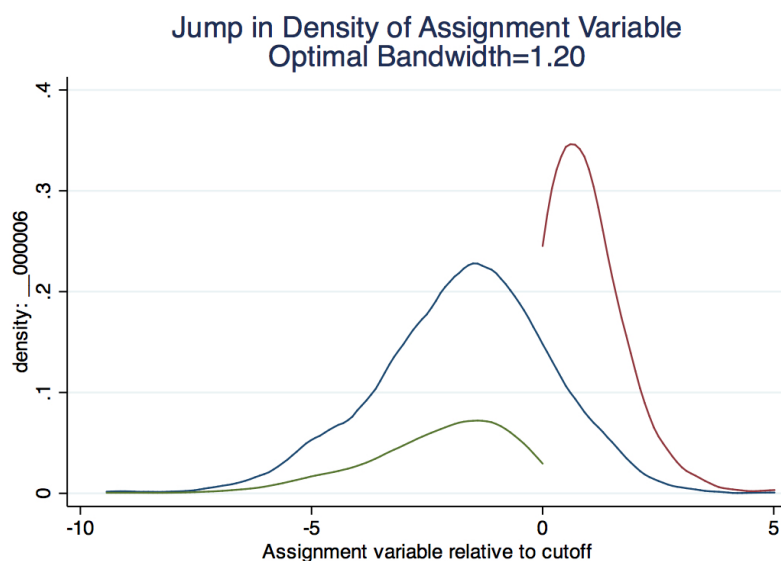


Figure 5 – *Jump in density of Assignment Variable for registration effective date March 30, 2012.* Figure 5 reports the results obtained by using the density test proposed by McCrary (2008). The results reported in Figure 5 support the presence of a discontinuity in March 2012 at the threshold of 150 million.

FIGURE 6

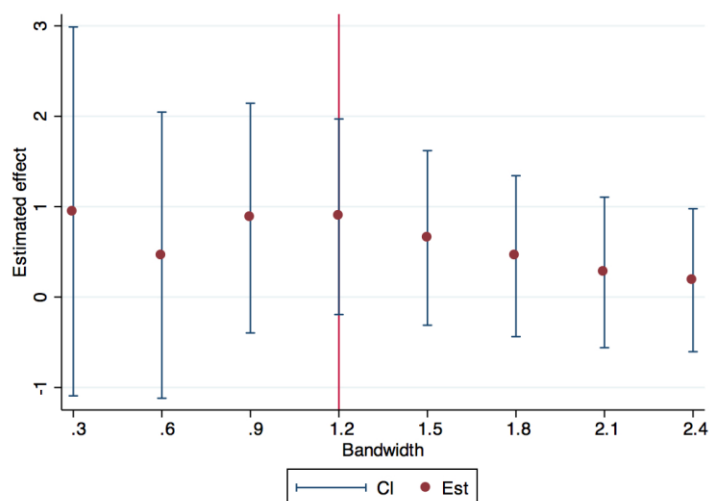


Figure 6 – *Confidence intervals for the estimated effect in March 30, 2012 when different bandwidths are applied.* Figure 6 shows that the estimates are statistically significant and rather stable also for larger bandwidths.

FIGURE 7

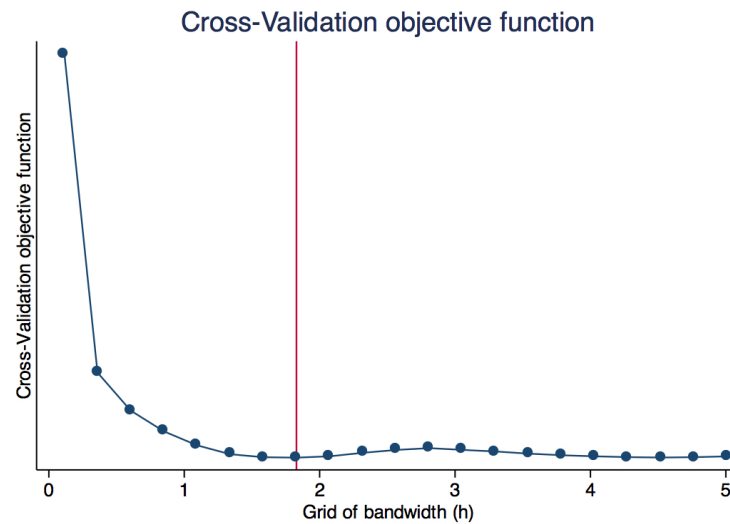


Figure 7 – *Confidence intervals for the estimated effect in March 30, 2012 for different bandwidths.* Figure 7 shows that the optimally chosen bandwidth is very close to the value that cross-validation would select, as implemented by Calonico et al. (2013), supporting the validity of the results. Increasing the bandwidth size does not dramatically affect our estimates.